

09/25/2026: Equivalence Relations

CSCI 246: Discrete Structures

Textbook reference: Sec 15, Scheinerman

Today's Agenda

- Quiz (≈ 15 mins)
- Mini-lecture (≈ 20 mins)
- Group exercises (≈ 15 mins)

Quiz logistics

- Please put your last name on the back.
- Please put your first and last name on the front.

Reminder

Please see me if you think your quiz was misgraded.

Weekly Quiz #4

1. (Sets.) How many subsets does the set $A = \{1, 2, 3\}$ have? Justify your answer.
2. (Quantifiers.) For each sentence below, write the negation of the sentence, but place the $\neg$ symbol as far to the right as possible. Then rewrite the negation in English.
 - a. $\forall x \in \mathbb{Z}, x$ is odd.
 - b. $\forall x \in \mathbb{Z}, x < 0$.
3. (Set Operations.) How many integers in the range 1 to 1000 (inclusive) are divisible by 2 or by 5? Justify your answer. *Hint:* Use the proposition giving the size of a union of sets.

Proposition: Size of a Union

Let A and B be sets. Then

$$|A \cup B| = |A| + |B| - |A \cap B|$$

Congruence mod n

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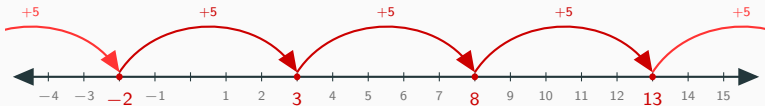
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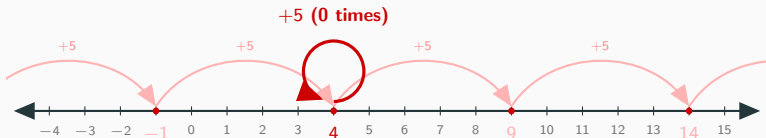
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$16 \equiv 3 \pmod{5}$? NO, because $16 - 3 = 13$ is not a multiple of 5

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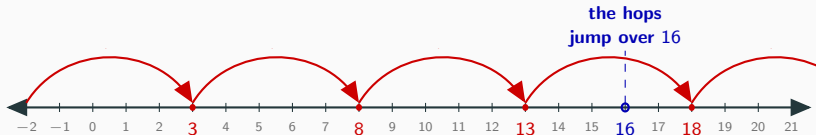
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Application: hashing features in machine learning

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A text model needs a **feature index** for every word. The vocabulary is unbounded, but the feature vector has **fixed** length m .

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Pick any hash function h (which converts the words to integers) and use $\text{index}(w) = h(w) \bmod m$. With $m = 5$:

word	$h(w)$	$h(w) \bmod 5$	bucket
"cat"	918273	3	[3]
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"the"	203948	3	[3] collision!
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The connection

"Same bucket" is **congruence mod m** : the m buckets are its equivalence classes, and a **collision** is two words in one class.

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- **No dictionary.** The mapping isn't stored anywhere; it's a formula. "cat" hashes to 918273, so bucket $918273 \bmod 5 = 3$, every time, on every machine. Zero bytes of lookup table.
- **Fixed memory.** You choose m up front and that's the whole model width. At $m = 2^{20}$ that's 4.2 MB of float32 weights, regardless of whether you've seen a thousand words or ten million.

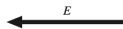
Application: Public-Key Cryptography



1

In private, Bob creates a public encryption function E and a secret decryption function D .

2



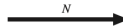
Bob sends his public encryption function E to Alice.

3

In private, Alice writes her message in ASCII, M . She uses Bob's function E to calculate $N = E(M)$.

4

Alice sends N to Bob.



5

In private, Bob uses his decryption function D to calculate $M = D(N)$. He now has Alice's message.

Eve sees E and N , but cannot calculate M from these.

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Definition (Scheinerman Def 14.1)

A **relation** is a set of ordered pairs.

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Let P be a set of people. Let C be a set of cities. Define the relation $R \subseteq P \times C$ by

$$R = \{(p, c) : \text{person } p \text{ has lived in city } c.\}$$

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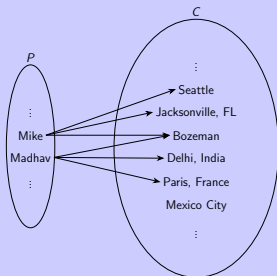
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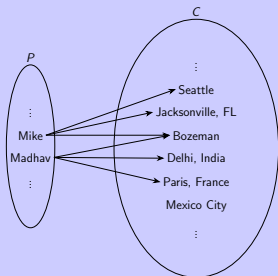
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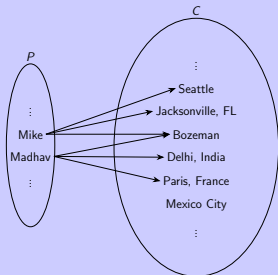
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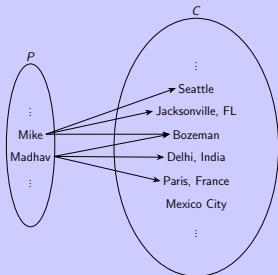
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- For a **function**, each input has **only one** output.
- For a **relation**, an input can have **more than one** output.

So a relation is a **generalization** of a function.

Review: Properties of Relations

Goal. Let's state the three main properties of relations and try to give an example/non-example for each.

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2. *Symmetry.* Sameness should go both ways. Otherwise you'd have a strange, one-sided notion of "sameness."
3. *Transitivity.* "Sameness" should spread consistently along chains of relations. (It would be weird if a was the same as b , and b was the same as c , but c wasn't the same as a .)

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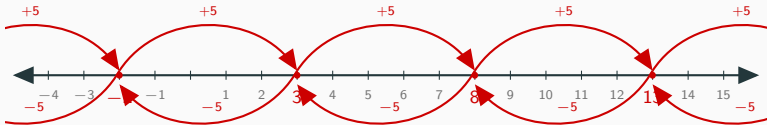
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- *Transitivity:* We need to check that if xRy and yRz , then xRz . That is, we need to check that if $n \mid (x - y)$ and $n \mid (y - z)$, then $n \mid (x - z)$.

Equivalence classes

Definition

Let R be an equivalence relation on a set A and let $a \in A$.

The **equivalence class** of a , denoted $[a]$, is the set of all elements of A related (by R) to a . That is,

$$[a] = \{x \in A : xRa\}$$

Example

The integers can be partitioned into the following equivalence classes

$$[0] = \{\dots, -6, -3, 0, 3, 6, \dots\}$$

$$[1] = \{\dots, -5, -2, 1, 4, 7, \dots\}$$

$$[2] = \{\dots, -4, -1, 2, 5, 8, \dots\}$$

under the relation of congruence mod 3.

Remark

Q: Why do equivalence relations matter?

A: They can make problem solving easier.

Example

Q: Why is $n^{70} - n^{22}$ always even?

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Now multiplication and addition is well-defined mod-2. (For justification, see Hampkins Ch2.) This means we can just use "0" and "1" as representatives in any multiplication or addition problem about odds and evens.

A: By the above POV, we only need to consider two cases:

$$n^{70} - n^{22} = 0^{70} - 0^{22} = 0 - 0 = 0 \quad \text{if } n = 0 \text{ (} n \text{ is even)}$$

$$n^{70} - n^{22} = 1^{70} - 1^{22} = 1 - 1 = 0 \quad \text{if } n = 1 \text{ (} n \text{ is odd)}$$

Either way, the result is 0 (the result is even).

Group exercises

Ashley, Isabelle: 7	Hager, Nathan: 6	Putnam, John: 3
Bailey, Tanna: 8	Horner, Jayden: 9	Ramos, Vydel: 11
Blaisdell, Olyver: 2	Johnson, Jaret: 4	Rathbun, Jedidiah: 2
Brown, Ashton: 11	Kintzel, Samantha: 13	Roberts, Matthew: 1
Brown, Jonathan: 5	Lameres, Alexis: 2	Roller, Nathan: 13
Cardenas, Dominic: 4	Lee, Holden: 3	Ryan, Otis: 10
Cutler, George: 8	Mayala, Reece: 5	Shevtsov, Timothy: 13
Fabik, Roland: 12	McLean, Connor: 6	Sinclair, Aidan: 3
Fellows, Audrey: 4	Mcbride, Daniel: 8	Smith, Charles: 7
Fitzgerald, Ryan: 10	Miller, Maverick: 12	Stensland, Emma: 9
Fuller, Ryan: 11	Moss, Emily: 5	Thiessen, Logan: 1
Griffen, Clara: 6	Mousseau, Matthew: 12	Thompson, Jaiden: 9
Grutsch, Tanner: 10	Neuman, Evan: 1	Vath, Ethan: 7

Group exercises

1. Which of the following are equivalence relations?
 - a. $|$ on $\mathbb{Z}$.
 - b. $\leq$ on $\mathbb{Z}$.
 - c. Is-an-anagram-of on the set of English words. (For instance, STOP is an anagram of POTS because we can form one from the other by rearranging its letters.)
 - d. $R = \{(1, 2), (2, 3), (3, 1)\}$ on the set $\{1, 2, 3\}$.
 - e. $\{1, 2, 3\} \times \{1, 2, 3\}$ on the set $\{1, 2, 3\}$.
 - f. $\{1, 2, 3\} \times \{1, 2, 3\}$ on the set $\{1, 2, 3, 4\}$.
2. For each equivalence relation below, find the requested equivalence class(es).
 - a. $R = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 4)\}$ on $\{1, 2, 3, 4\}$. Find $[1], [2], [3], [4]$
 - b. R is has-the-same-tens-digit on the set $\{x \in \mathbb{Z} : 100 < x < 200\}$. Find $[123]$.
 - c. R is has-the-same-parents-as on the set of all human beings. Find $[\text{you}]$.
 - d. R is has-the-same-size-as on $2^{\{1, 2, 3, 4, 5\}}$. Find $[\{1, 3\}]$.
3. Let n be a positive integer. Prove that congruence modulo n is an equivalence relation on the set of integers.

Solution to group exercise #1

Problem. Which of the following are equivalence relations?

- a. $|$ on $\mathbb{Z}$.
- b. $\leq$ on $\mathbb{Z}$.
- c. Is-an-anagram-of on the set of English words. (For instance, STOP is an anagram of POTS because we can form one from the other by rearranging its letters.)
- d. $R = \{(1, 2), (2, 3), (3, 1)\}$ on the set $\{1, 2, 3\}$.
- e. $\{1, 2, 3\} \times \{1, 2, 3\}$ on the set $\{1, 2, 3\}$.
- f. $\{1, 2, 3\} \times \{1, 2, 3\}$ on the set $\{1, 2, 3, 4\}$.

Solution.

- a. No, because the relation is not symmetric. For example, $3R6$ but $6 \not R 3$.
- b. No, because the relation is not symmetric. For example, $3R4$ but $4 \not R 3$.
- c. Yes.
- d. No, because the relation is not transitive. In particular, we have $1R2$ and $2R3$, but $1 \not R 3$.
- e. Yes.
- f. No, because the relation is not reflexive. In particular, $4 \not R 4$.

Solution to group exercise #2

Problem. For each equivalence relation below, find the requested equivalence class(es).

- a. $R = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 4)\}$ on $\{1, 2, 3, 4\}$. Find $[1], [2], [3], [4]$
- b. R is has-the-same-tens-digit on the set $\{x \in \mathbb{Z} : 100 < x < 200\}$. Find $[123]$.
- c. R is has-the-same-parents-as on the set of all human beings. Find $[you]$.
- d. R is has-the-same-size-as on $2^{\{1,2,3,4,5\}}$. Find $[\{1, 3\}]$.

Solution.

- a. We have $[1] = [2] = \{1, 2\}$, $[3] = \{3\}$, and $[4] = \{4\}$.
- b. We have $[123] = \{120, 121, 122, 123, 124, 125, 126, 127, 128, 129\}$.
- c. The answer depends on who's answering. I have one sister named Rachael, so for me, $[me] = \{me, Rachael\}$.
- d. We have

$$[\{1, 3\}] = \left\{ \{1, 2\}, \{1, 3\}, \{1, 4\}, \{1, 5\}, \{2, 3\}, \{2, 4\}, \{2, 5\}, \{3, 4\}, \{3, 5\}, \{4, 5\} \right\}.$$

Solution to exercise #3 (proving an equivalence relation)

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We verify the three properties of an equivalence relationship below.

- *Reflexivity:*

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- *Reflexivity*: We need to check xRx . That is, we need to check $n|(x - x)$. In other words, we need to check $n|0$. By definition of divisibility, we need to check that there is an integer c such that $nc = 0$. This is satisfied by setting $c = 0$. ✓
- *Symmetry*:

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- *Symmetry*: We need to check that if xRy , then yRx . In other words, we need to check that if $n|(x - y)$, then $n|(y - x)$. Let $n|(x - y)$. Then there is an integer c such that $nc = x - y$. Hence $n(-c) = y - x$. So $n|(y - x)$. ✓
- *Transitivity*:

Solution to exercise #3 (proving an equivalence relation)

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- *Reflexivity*: We need to check xRx . That is, we need to check $n|(x - x)$. In other words, we need to check $n|0$. By definition of divisibility, we need to check that there is an integer c such that $nc = 0$. This is satisfied by setting $c = 0$. ✓
- *Symmetry*: We need to check that if xRy , then yRx . In other words, we need to check that if $n|(x - y)$, then $n|(y - x)$. Let $n|(x - y)$. Then there is an integer c such that $nc = x - y$. Hence $n(-c) = y - x$. So $n|(y - x)$. ✓
- *Transitivity*: We need to check that if xRy and yRz , then xRz . That is, we need to check that if $n|(x - y)$ and $n|(y - z)$, then $n|(x - z)$. By assumption, there are integers c and d such that $nc = (x - y)$ and $nd = (y - z)$. Now we write

$$x - z = (x - y) + (y - z) = nc + nd = n(c + d).$$

Since $c + d$ is an integer, clearly $n|x - z$. ✓